

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination
May 2011

CL 209 Structural Analysis I

Date: 11.05.2011, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a)** Classify the frame and truss given in **Figure 1 to 3** as statically determinate, statically indeterminate or unstable. If statically indeterminate, specify the degrees of indeterminacy. **03**
- (b)** Determine Degrees of Kinematic Indeterminacy of the beam given in **Figure 4 to 6**. **03**
- Q - 2 (a)** Show that in case of a thin cylindrical shell subjected to an internal fluid pressure, the circumferential stress is twice the longitudinal stress. **04**
- (b)** A thin spherical pressure vessel of diameter 1.5 m, thickness of metal 15 mm and the efficiency of longitudinal joint is 70%. If the maximum tensile stress of the plate is 80 N/mm², calculate the permissible steam pressure in the vessel. Calculate also the circumferential stress in the solid plate section and longitudinal stress in the plate section through the joint if efficiency of circumferential joint is 60%. **05**
- (c)** A simply supported beam of uniform flexural rigidity EI and span L carries two symmetrically placed loads P at one-third of the span from each end. Find the expression for deflection at the middle of the span. **05**
- (d)** A simply supported R.C. rectangular beam of span 4 m carrying a UDL of 10 kN/m over the entire span and a point load of 50 kN at a distance of 2 m from the left end. The cross-section is 150 mm* 300 mm. Find the maximum slope and deflection of the beam. Take $E = 2 \times 10^4$ N/mm². **05**

OR

- Q - 2 (a)** A watermain 500 mm diameter contains water at a pressure head of 100 m. If the weight of water is 10 kN/m³ and allowable stress in the pipe 20 N/mm², calculate the thickness of the pipe. **04**
- (b)** A cylindrical shell 1.5 m long, internal diameter 300 mm, and wall thickness 10 mm is filled up with a fluid at atmospheric pressure. If additional of 300 c.c. is pumped in the shell, find the pressure exerted by the fluid on the shell. Take $E = 2 \times 10^5$ N/mm² and $l/m = 0.3$ and the hoop stress induced. **05**
- (c)** A simply supported R.C. rectangular beam of span 3 m is carrying a UDL of 8 kN/m over the entire span. If the allowable bending stress is 7 N/mm² and the allowable deflection is 10 mm, find the necessary width and depth of the section. Take $E = 1 \times 10^4$ N/mm². **05**
- (d)** A simply supported beam of span 4 m and circular cross-section having diameter 200 mm is **05**

loaded as shown in **Figure 7**. Find the maximum deflection of the beam.

$$E = 2 \times 10^4 \text{ N/mm}^2.$$

Q - 3 Attempt any Two

10

- (a) Derive Euler's formula for the critical load of a column having one end fixed and other end free.
- (b) A steel bar of solid circular cross-section is 50 mm in diameter. The bar is pinned at both ends and subjected to axial compression. If the limit of proportionality of the material is 210 MPa & $E = 200 \text{ GPa}$. Determine the minimum length for which Euler's formula is valid. Also determine the value of Euler's buckling load if the column has this minimum length.
- (c) A stanchion, 6 meters long, ends free, is built-up of two 400 mm x 100 mm standard channels placed 150 mm back to back, with one 400 mm x 12 mm plate welded to each flange. It carries a load of 1300 kN, which is off y-axis, in the vertical plane through x-axis. Calculate the permissible eccentricity if the maximum permissible compressive stress is 105 N/mm^2 . For each channel, area of section = 62.93 cm^2 , distance of centroid from back = 2.42 cm, $I_x = 15082.6 \text{ cm}^4$, $I_y = 504.8 \text{ cm}^4$, $E = 2 \times 10^5 \text{ N/mm}^2$.

SECTION - II

Q - 4 (a) What do you mean by the term limit of eccentricity and core of a column section? Find the limit of eccentricity and core for a solid circular section of diameter D. 03

(b) A Steel pillar of circular cross-section is of external diameter 40 cm and metal thickness 4 cm. A projecting bracket attached to it carries a load of 10 tonnes. The load line being off the axis of the pillar by 25 cm. Compute the maximum and minimum intensities of stress in the section. 04

(c) A hollow rectangular column of external and Internal dimension is 2.4 m X 1.8 m and 1.2 m X 1.2 m respectively. Calculate the safe load that can be placed at an eccentricity of 50 cm on a plane bisecting the longer side, if the maximum compressive stress is not to exceed 50 kg/cm^2 . 04

Q - 5 Attempt any Two

12

(a) A simply supported beam AB has a span of 5 m. Draw influence lines for R_A , R_B , V_x and M_x for a section X at 2m from left hand support.

Also find values of R_A , R_B , V_x , M_x when a series of four concentrated loads 10 kN, 12 kN, 15 kN and 8 kN are acting at 1m, 2m, 3m and 4m from A.

(b) Two wheel loads 12 kN and 20 kN with fixed distance 2m between them and 12 kN load leading, crosses a beam of span 8m from left to right.

Draw the influence line for shear force and bending moment for a point 3 m from left support and find maximum values of shear force and bending moment.

(c) An UDL of 12 kN/m and 3m length crosses a simply supported girder of span 10 m from left to right. Draw influence line for S.F. and B.M. at section 4 m from left and find maximum S.F. and B.M. at this section.

Q - 6 Attempt any Two

12

(a) A three hinged parabolic arch having supports at different levels as shown in **Figure 8**,

carries an UDL of intensity 30 kN/m over the left portion of the crown. Determine the horizontal thrust developed. Find also the bending moment, normal thrust, radial shear force developed at a section 15 m from the left support.

- (b) A three hinged circular arch consists of a portion AC of a radius 3 m and rise of the hinge C with respect to the left abutment is 3 m. The right hand portion CB is of radius 8 m and the horizontal distance BC is 7 m. If a concentrated load of 10 kN acts at 6 m from the left hand end, determine the reactions at the hinges and maximum bending moment on the arch.
- (c) A parabolic three hinged arch carries a UDL of 30 kN/m on the left of the span. It has a span of 16 m and a central rise of 3 m. Determine the resultant reactions at supports. Find the bending moments, normal thrust and radial shear at section, 2 m from left support.

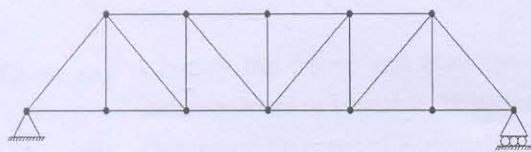


Fig. 1

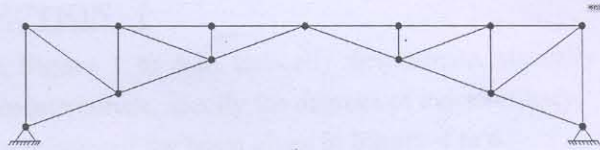


Fig. 2

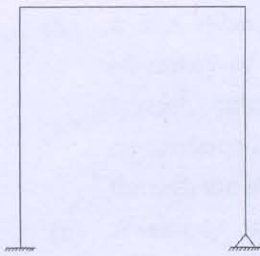


Fig. 3

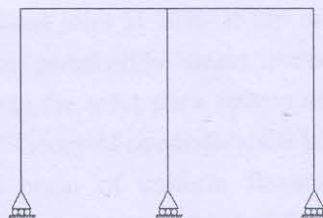


Fig. 4

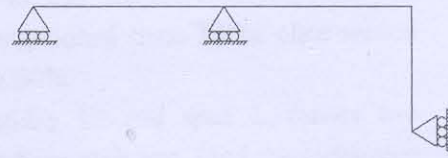


Fig. 5

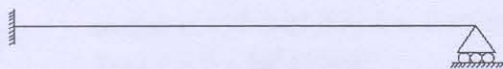


Fig. 6

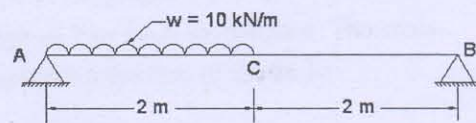


Fig. 7

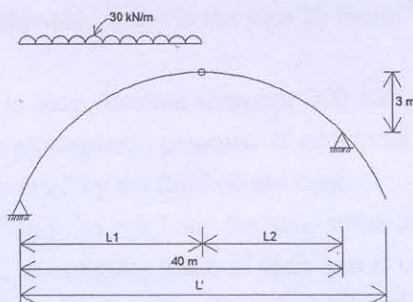


Fig. 8
